



Driver Fatigue Detection & Progression System

An Advanced Deep Learning Computer Vision Framework for Real-Time Fleet Risk Assessment

ADAS

FLEET SAFETY

COMPUTER VISION

The Operational Challenge

Why It Matters

Driver drowsiness is a leading contributor to global highway logistics fatalities. Raw image streaming demands **instant, low-latency identification** before a threat escalates into an incident.

Core Visual Concepts

→ Eye State Analysis

Detecting structural shifts across Open vs. Closed positions.

→ Mouth State Analysis

Segmenting facial geometry into Yawn vs. No Yawn categories.

→ System Fusion Core

Concurrent stream fusion to build an early warning system.

Dataset Partitioning Strategy

1,000 unified high-resolution driver cabin images, partitioned to ensure unbiased evaluation and robust generalisation.

Training Set

800 images — Used for model backpropagation and structural weight optimisation.

Validation Set

100 images — Dedicated to hyperparameter tuning, epoch monitoring, and early stopping.

Test Set

100 images — Held completely isolated for a final, unbiased accuracy benchmark.

Data Augmentation & Balance Control

Prior to feature extraction, state distributions are evaluated. If spatial or numerical class imbalances exist, a dynamic augmentation pipeline is initialised via `tf.keras.Sequential` layers.



Stochastic Rotation

Applies up to $\pm 15^\circ$ variations to handle natural driver head tilts.



Intrinsic Zooming

Modifies scale by $\pm 10\%$ to replicate seating distance offsets.



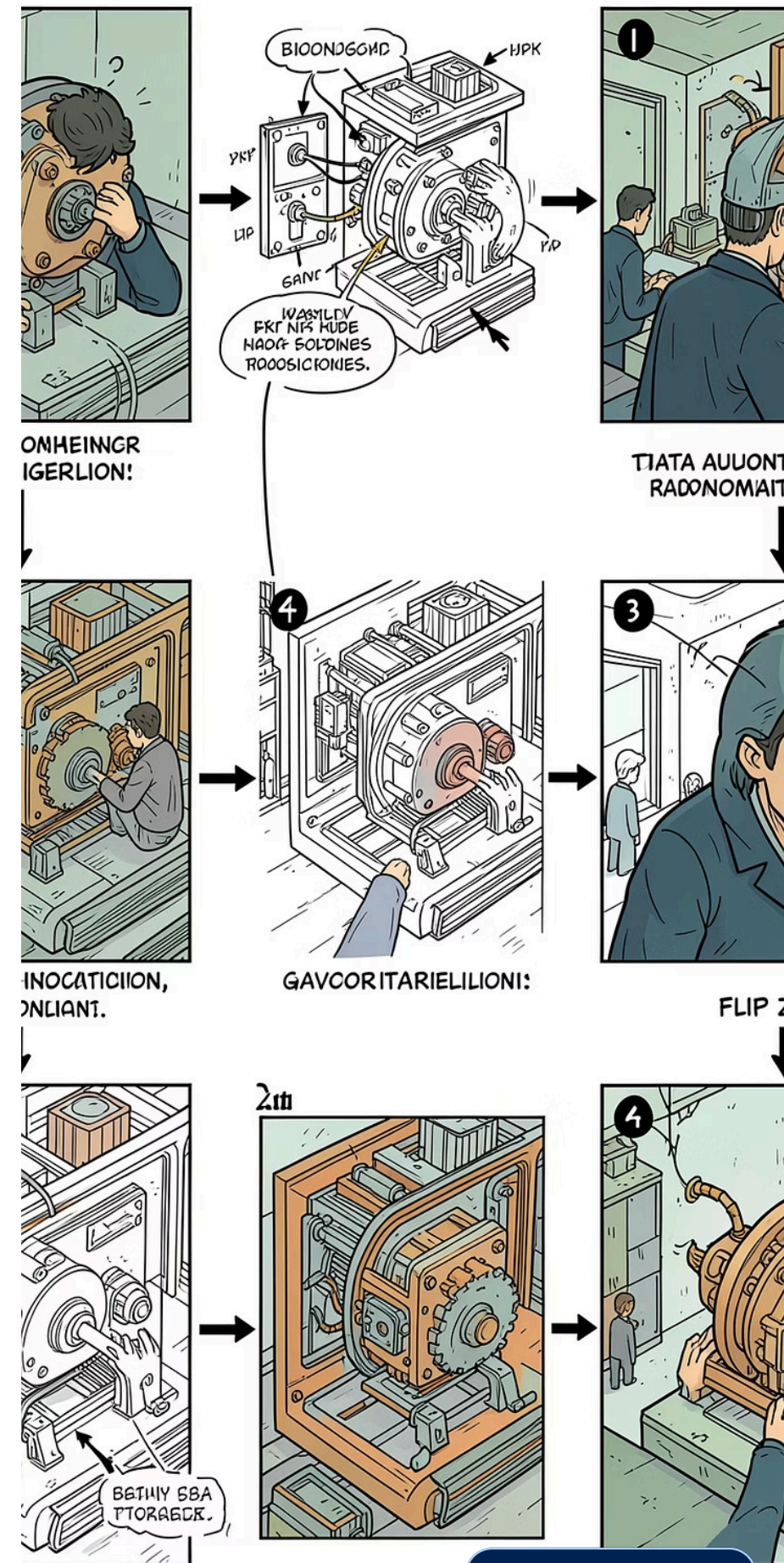
Exposure Adjustment

Modifies pixel intensity maps for robustness against day-to-night lighting transitions.

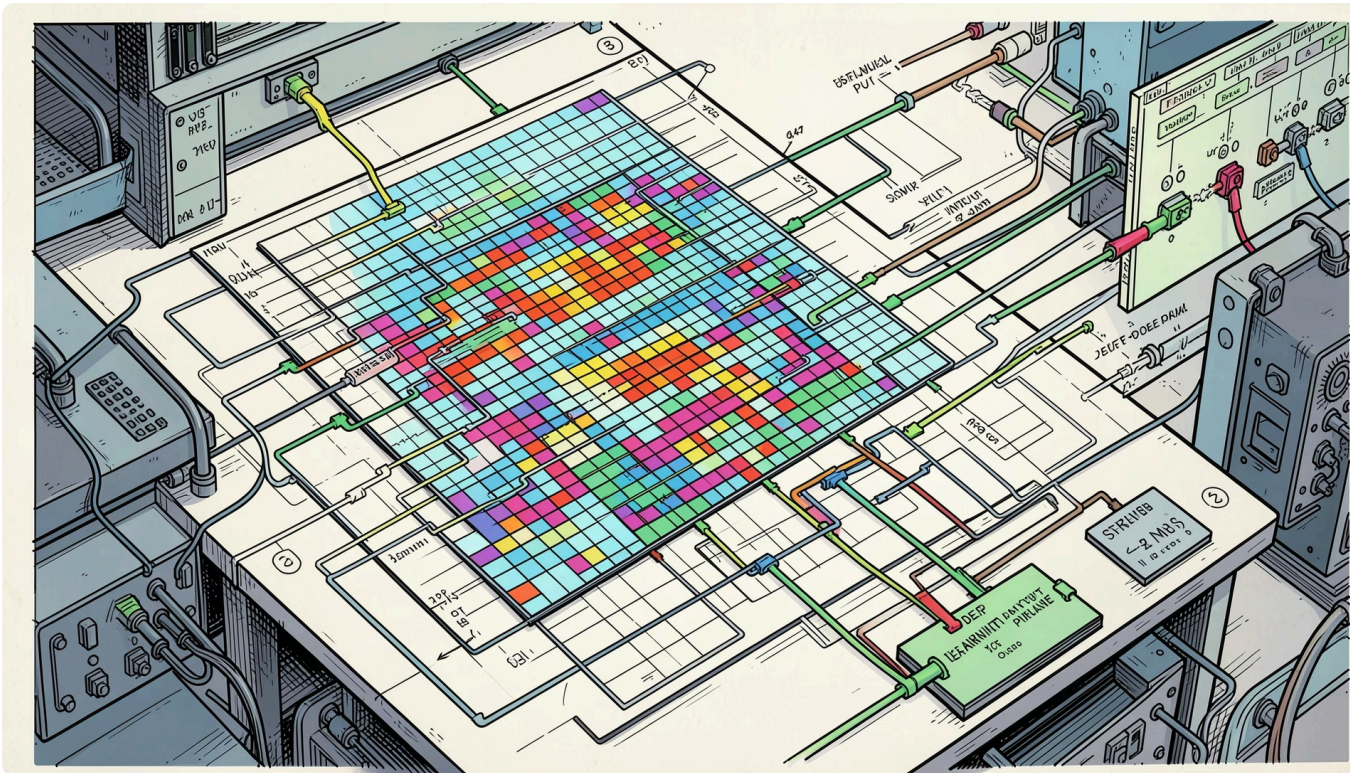


Horizontal Flipping

Enforces mirror invariance for both left- and right-hand drive configurations.



Core Preprocessing Pipeline



Standardising Tensor Stream Formats

Step 1 – Dimensional Resizing

All incoming images are
downsampled to **224 × 224 pixels**,
ensuring exact matrix alignment with
minimal compute latency.

Step 2 — Floating Point Normalisation

Native 8-bit colour matrices within **[0, 255]** are transformed to a clean **[0.0, 1.0]** float range via min-max scaling:

$$X_{\text{norm}} = X / 255.0$$

Model 1: Custom CNN Architecture

EDGE-OPTIMISED

BUILT FROM SCRATCH

Architecture Stack

Input (224×224×3)

→ Conv2D (32, 3×3) → MaxPool
→ Conv2D (64, 3×3) → MaxPool
→ Conv2D (128, 3×3) → MaxPool
→ Flatten → Dense (128)
→ Dropout (0.5) → Output Head

Design Rationale

1 Feature Deepening

Expands kernel space depth while compressing spatial dimensions across three convolutional blocks.

2 Overfitting Isolation

High dropout (0.5) maintains strict layer validation stability during training.

3 Edge Deployment

Engineered to minimise footprint on low-power edge processing hardware.

Model 2: Transfer Learning — MobileNetV2

Leverages rich architectural filters pre-trained on ImageNet, adapted for custom driver facial classification.



Phase I — Weight Isolation

Inverted residual blocks are **frozen** to retain deep global edge-detection features.



Phase II — Head Adaptation

Global average pooling followed by dense output mappings for custom driver facial variables.



Phase III — Fine-Tuning

Upper residual blocks unfrozen with a conservative learning rate $\alpha = 0.0001$ for accuracy optimisation.

Decision Fusion Logic — 3-Level Fatigue Model

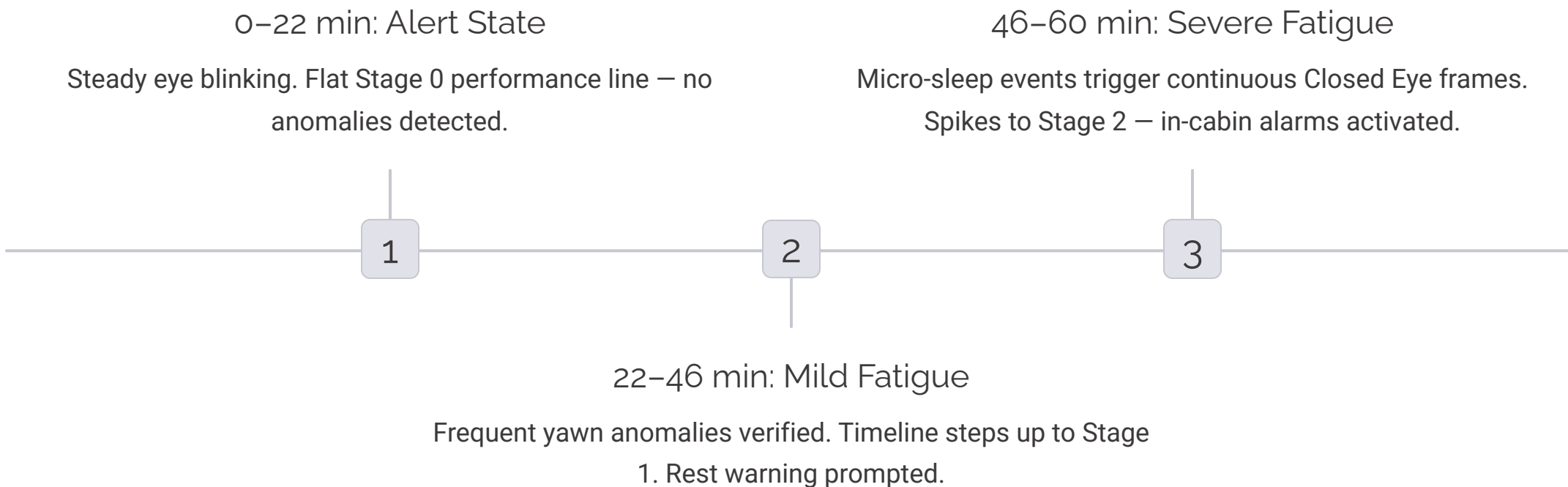
Independent classifier outputs are consolidated into a unified risk management profile.

Stage	Fatigue Level	Algorithmic Condition
Stage 0	Alert	Eye predicted Open AND Mouth predicted No Yawn
Stage 1	Mild Fatigue	Mouth predicted Yawn (regardless of eye state — early warning)
Stage 2	Severe Fatigue	Eye predicted Closed — triggers immediate high-priority system alarms

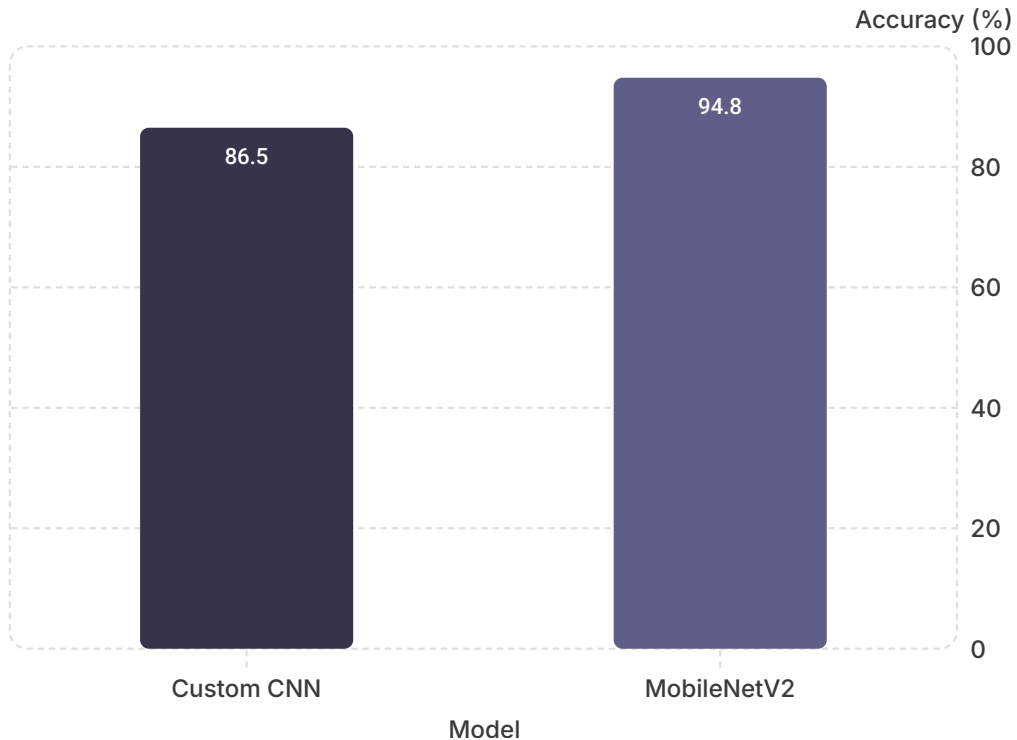
 **Critical Safety Override:** If Eye Closed state is active, the system bypasses all mouth status pipelines and escalates immediately to Stage 2.

Driver Fatigue Progression Curve

Temporal assessment over a simulated 60-minute driving profile, using 1-minute frame batch windows.



Performance Benchmarking & Recommendations



Architectural Trade-Offs

Future Recommendations

- **Cross-Spectral Data:** Integrate IR bands to eliminate nighttime classification errors.
- **Temporal Recurrence:** Introduce LSTM blocks or Kalman smoothing to filter frame prediction noise and reduce false alarms.

MobileNetV2

94.8% accuracy — superior generalisation under uneven lighting. Latency: ~18 ms.

Custom CNN

86.5% accuracy — ultra-low ~8 ms latency, ideal for low-power edge units.